

Connecting Two Routers in Packet Tracer

<https://gemini.google.com/app/1695944fd36e8664>

User prompt: How to connect these 2 routers

Response: To link **Router0** and **Router1** in Cisco Packet Tracer and get traffic flowing end-to-end, follow these steps:

Step 1: Physical Cable Connection

- Select the **Connections** lightning bolt tool from the bottom toolbar.
- Choose **Copper Crossover** (or **Copper Straight-Through**, as modern Packet Tracer 1941 routers auto-MDIX).
- Click on **Router0** and choose an available Ethernet port (e.g., GigabitEthernet0/1 or GigabitEthernet0/0 if unused).
- Click on **Router1** and connect to an available Ethernet port (e.g., GigabitEthernet0/1).

(Alternatively, if you run out of Ethernet ports, double-click each router, turn off the power switch, insert a HWIC/WIC module like **HWIC-2T**, turn power back on, and connect them using a **Serial DCE cable**).

Step 2: Assign IP Addresses to Inter-Router Interfaces Pick a dedicated subnet for the point-to-point link between the routers (e.g., 10.0.0.0/30 or 192.168.100.0/24).

• Router0 Configuration:

```
Router0> enable
Router0# configure terminal
Router0(config)# interface g0/1
Router0(config-if)# ip address 10.0.0.1 255.255.255.252
Router0(config-if)# no shutdown
```

• Router1 Configuration:

```
Router1> enable
Router1# configure terminal
Router1(config)# interface g0/1
Router1(config-if)# ip address 10.0.0.2 255.255.255.252
Router1(config-if)# no shutdown
```

Step 3: Configure Routing To allow PCs on the left side to ping PCs on the right side, add static routes (or enable OSPF/RIP).

• Static Route on Router0 (pointing to Router1's internal subnets):

```
Router0(config)# ip route <destination_subnet_on_right> <subnet_mask> 10.0.0.2
```

• Static Route on Router1 (pointing to Router0's internal subnets):

```
Router1(config)# ip route <destination_subnet_on_left> <subnet_mask> 10.0.0.1
```

(Or set a default route `ip route 0.0.0.0 0.0.0.0 10.0.0.x` on both ends if these are simple gateway connections).

User prompt: well theres no available ethernet port on either of these routers

Response: Since the two built-in GigabitEthernet ports on the Cisco 1941 are already taken up by your switches, you need to add an expansion module to both routers in Packet Tracer.

Here are the two ways to add ports:

Option 1: Add Serial Ports (Standard Cisco Lab Setup)

This is the recommended method for router-to-router links in Packet Tracer.

1. Click on **Router0** to open its window and make sure you are in the **Physical** tab.
2. Turn **OFF** the router using the power switch on the router image.
3. From the **Modules** list on the left, select **HWIC-2T**.
4. Drag the **HWIC-2T** card into the empty slot on the right side of the router.
5. Turn the router back **ON**.
6. Repeat steps 1–5 for **Router1**.
7. In the bottom connections menu, select the **Serial DCE cable** (the red wire with a clock icon).
8. Click **Router0**, select Serial0/0/0, then click **Router1** and select Serial0/0/0.

Option 2: Add Extra Ethernet Ports (If you prefer Ethernet)

1. Click on **Router0** → **Physical** tab.
2. Turn **OFF** the router using the power switch.
3. From the **Modules** list, select **HWIC-4ESW** (adds 4 Switchports) or **HWIC-1GE-SFP-CU** (adds a Gigabit Ethernet port).
4. Drag the module into an empty slot on the router.
5. Turn the router back **ON**.
6. Repeat steps 1–5 for **Router1**.
7. Connect the routers using a **Copper Crossover** (or **Copper Straight-Through**) cable using the newly added interface ports (e.g., FastEthernet0/0/0).

User prompt: Well I did this Yes — I can see the problem from your topology. Your IP addressing doesn't match the physical connections in the diagram. The main mistake in your diagram: Router0 has two GigabitEthernet connections: G0/0 → Switch0 G0/1 → Switch1 Router1 has two GigabitEthernet connections: G0/0 → Switch2 G0/1 → Switch3 The dashed link between Router0 and Router1 is the router-to-router link, which should use the 10.0.0.0/30 network. But your current configuration puts 10.0.0.1 on Router0 G0/1, even though G0/1 is connected to Switch1. So let's configure the network properly without needing to connect Switch0 ↔ Switch1 or Switch2 ↔ Switch3. 1. Correct network design We'll use four separate LANs: 10.0.0.0/30 Router0 ===== Router1 /\ /\ /\ LAN 10 LAN 11 LAN 20 LAN 21 /\ /\ Switch0 Switch1 Switch2 Switch3 /\ /\ /\ PC0 PC1 PC2 PC3 PC4 PC5 PC6 PC7 IP networks

NetworkPurpose 192.168.10.0/24 Switch0 192.168.11.0/24 Switch1 192.168.20.0/24 Switch2 192.168.21.0/24 Switch3 10.0.0.0/30 Router0 ↔ Router1 2. Router0 configuration Router0 Interface IPv4 Address Subnet Mask Connected to G0/0 192.168.10.1 255.255.255.0 Switch0 G0/1 192.168.11.1 255.255.255.0 Switch1 Serial interface 10.0.0.1 255.255.255.0 Router1 Important: The 10.0.0.1 address should NOT be on G0/1. It belongs on the serial interface connected to Router1. 3. Router1 configuration Router1 Interface IPv4 Address Subnet Mask Connected to G0/0 192.168.20.1 255.255.255.0 Switch2 G0/1 192.168.21.1 255.255.255.0 Switch3 Serial interface 10.0.0.2 255.255.255.0 Router0 4. PC configurations Now each PC gets an address from the network of the switch it is connected to. Switch0 → Router0 G0/0 PC IPv4 Address Subnet Mask Default Gateway PC0 192.168.10.2 255.255.255.0 192.168.10.1 PC1 192.168.10.3 255.255.255.0 192.168.10.1 Switch1 → Router0 G0/1 PC IPv4 Address Subnet Mask Default Gateway PC2 192.168.11.2 255.255.255.0 192.168.11.1 PC3 192.168.11.3 255.255.255.0 192.168.11.1 Switch2 → Router1 G0/0 PC IPv4 Address Subnet Mask Default Gateway PC4 192.168.20.2 255.255.255.0 192.168.20.1 PC5 192.168.20.3 255.255.255.0 192.168.20.1 Switch3 → Router1 G0/1 PC IPv4 Address Subnet Mask Default Gateway PC6 192.168.21.2 255.255.255.0 192.168.21.1 PC7 192.168.21.3 255.255.255.0 192.168.21.1 5. Now you need routing Even with the correct IP addresses, Router0 doesn't automatically know about Router1's LANs, and vice versa. The easiest solution for this small topology is static routing. Router0 Enter: Router> enable Router# configure terminal Router(config)# interface gigabitEthernet 0/0 Router(config-if)# ip address 192.168.10.1 255.255.255.0 Router(config-if)# no shutdown Router(config-if)# exit Router(config)# interface gigabitEthernet 0/1 Router(config-if)# ip address 192.168.11.1 255.255.255.0 Router(config-if)# no shutdown Router(config-if)# exit Router(config)# interface serial 0/0/0 Router(config-if)# ip address 10.0.0.1 255.255.255.0 Router(config-if)# no shutdown Router(config-if)# exit Your serial interface might be Serial0/0/1 instead. Use whichever interface is actually connected by the dashed line. Then add routes: Router(config)# ip route 192.168.20.0 255.255.255.0 10.0.0.2 Router(config)# ip route 192.168.21.0 255.255.255.0 10.0.0.2 6. Router1 Router> enable Router# configure terminal Router(config)# interface gigabitEthernet 0/0 Router(config-if)# ip address 192.168.20.1 255.255.255.0 Router(config-if)# no shutdown Router(config-if)# exit Router(config)# interface gigabitEthernet 0/1 Router(config-if)# ip address 192.168.21.1 255.255.255.0 Router(config-if)# no shutdown Router(config-if)# exit Router(config)# interface serial 0/0/0 Router(config-if)# ip address 10.0.0.2 255.255.255.0 Router(config-if)# no shutdown Router(config-if)# exit Then: Router(config)# ip route 192.168.10.0 255.255.255.0 10.0.0.1 Router(config)# ip route 192.168.11.0 255.255.255.0 10.0.0.1 7. If the serial link is DCE One side of a serial connection in Packet Tracer is usually DCE. On the DCE router, configure: Router(config)# interface serial 0/0/0 Router(config-if)# clock rate 64000 You can check which side is DCE with: show controllers serial 0/0/0 Only the DCE side needs the clock rate. 8. Final configuration Your complete addressing should therefore be: Device Interface IP Address Mask Router0 G0/0 192.168.10.1 255.255.255.0 Router0 G0/1 192.168.11.1 255.255.255.0 Router0 Serial 10.0.0.1 255.255.255.0 Router1 G0/0 192.168.20.1 255.255.255.0 Router1 G0/1 192.168.21.1 255.255.255.0 Router1 Serial 10.0.0.2 255.255.255.0 PC0 192.168.10.2 255.255.255.0 PC1 192.168.10.3 255.255.255.0 PC2 192.168.11.2 255.255.255.0 PC3 192.168.11.3 255.255.255.0 PC4 192.168.20.2 255.255.255.0 PC5 192.168.20.3 255.255.255.0 PC6 192.168.21.2 255.255.255.0 PC7 192.168.21.3 255.255.255.0 And: PC IP Mask Gateway PC0 192.168.10.2 255.255.255.0 192.168.10.1 PC1 192.168.10.3 255.255.255.0 192.168.10.1 PC2 192.168.11.2 255.255.255.0 192.168.11.1 PC3 192.168.11.3 255.255.255.0 192.168.11.1 PC4 192.168.20.2 255.255.255.0 192.168.20.1 PC5 192.168.20.3 255.255.255.0 192.168.20.1 PC6 192.168.21.2 255.255.255.0 192.168.21.1 PC7 192.168.21.3 255.255.255.0 192.168.21.1 Why your current setup fails Your current setup effectively does this: Router0 — G0/0 → 192.168.10.0/24 → Switch0 → PC0/PC1 — G0/1 → 10.0.0.0/30 → Switch1 → PC2/PC3 ✗ PC2 and PC3 are connected to a switch, but their gateway would need to be 192.168.11.1, not 10.0.0.1. The 10.0.0.0/30 network should be exclusively Router0 ↔ Router1. With the corrected setup and static routes, PC0 can ping PC7, PC3 can ping PC5, PC1 can ping PC6, etc. — all 8 PCs can communicate with each other. But PC0 and 1 can now communicate with PC 2 and PC 3 but not with the other side I added clock rate for both routers to be 64000

Response: If PC0/PC1 can talk to PC2/PC3, your local interfaces and gateways on Router0 are configured properly. The issue with reaching the right side (Router1's networks) usually stems from **missing reverse routes**, **subnet mask mismatches**, or **ICMP ARP timeouts** during the first ping in Packet Tracer.

Here is how to isolate and fix it step-by-step:

1. Test Router-to-Router Reachability First, check if the two routers can actually talk across the red serial line.

- Open CLI on **Router0** and run: `ping 10.0.0.2`
- If it fails (0% success rate), the serial link itself is down:
 - Run `show ip interface brief` on both routers to verify Serial0/0/0 (or Serial0/0/1) shows **Status: UP** and **Protocol: UP**.
 - Make sure both ends use mask 255.255.255.252.

2. Check Static Routes on BOTH Routers A very common issue is configuring routes on Router0 to reach Router1, but forgetting to tell Router1 how to get back to Router0. Routing must be bi-directional.

- On **Router0**, run `show ip route` in enable mode. You must see:
 - S 192.168.20.0/24 [1/0] via 10.0.0.2
 - S 192.168.21.0/24 [1/0] via 10.0.0.2
 - (If missing, re-add: `ip route 192.168.20.0 255.255.255.0 10.0.0.2` and `ip route 192.168.21.0 255.255.255.0 10.0.0.2`)
- On **Router1**, run `show ip route`. You must see:
 - S 192.168.10.0/24 [1/0] via 10.0.0.1
 - S 192.168.11.0/24 [1/0] via 10.0.0.1
 - (If missing, re-add: `ip route 192.168.10.0 255.255.255.0 10.0.0.1` and `ip route 192.168.11.0 255.255.255.0 10.0.0.1`)

3. Packet Tracer Initial Ping Timeout In Packet Tracer, the very first ping from PC0 to PC4 almost always times out while ARP populates across multiple hops.

- In the PC0 Command Prompt, issue the ping **twice** in a row (ping 192.168.20.2).
- If the first attempt gives Request timed out on the first 1-2 packets but replies afterwards, it is working as intended.

4. Quick Fix Alternative: Default Routes If you want to simplify static routing without listing every single LAN subnet manually:

- On **Router0**:

```
Router0(config)# ip route 0.0.0.0 0.0.0.0 10.0.0.2
```
- On **Router1**:

```
Router1(config)# ip route 0.0.0.0 0.0.0.0 10.0.0.1
```

This forces any non-local traffic automatically across the serial link.

Response: You got the % Invalid input detected error because you are inside **interface configuration mode** (Router(config-if)#). Cisco IOS commands like ping can only be run directly from **Privileged EXEC mode** (Router#).

Also, notice from your screenshot that your serial interface is **Serial0/1/0** (not Serial0/0/0).

Here is how to run your commands correctly:

1. Fix the Command Modes

Type `do` before ping to run it while inside interface mode, or `exit` back to Privileged EXEC mode:

Option A (Using do from current mode):

```
Router(config-if)# do ping 10.0.0.2
```

Option B (Exiting back to Router#):

```
Router(config-if)# end
Router# ping 10.0.0.2
```

2. Complete IP & Routing Configuration for Serial0/1/0

Since your serial port is Serial0/1/0, apply the full IP setup on both routers:

On Router0:

```
Router0> enable
Router0# configure terminal
Router0(config)# interface Serial0/1/0
Router0(config-if)# ip address 10.0.0.1 255.255.255.252
Router0(config-if)# clock rate 64000
Router0(config-if)# no shutdown
Router0(config-if)# exit
Router0(config)# ip route 192.168.20.0 255.255.255.0 10.0.0.2
Router0(config)# ip route 192.168.21.0 255.255.255.0 10.0.0.2
```

On Router1:

```
Router1> enable
Router1# configure terminal
Router1(config)# interface Serial0/1/0
```

```
Router1(config-if)# ip address 10.0.0.2 255.255.255.252
Router1(config-if)# no shutdown
Router1(config-if)# exit
Router1(config)# ip route 192.168.10.0 255.255.255.0 10.0.0.1
Router1(config)# ip route 192.168.11.0 255.255.255.0 10.0.0.1
```

Once applied, type `do ping 10.0.0.2` on Router0 to verify the link succeeds.

Response: That 100% success rate confirms the serial link between Router0 and Router1 is fully up and working!

Now, to get PC0 (or any PC on the left) to reach PC4 (or any PC on the right), double-check these final two configurations:

1. Ensure Static Routes Are Applied

Run `do show ip route` on both routers to verify that each router knows how to reach the other side's networks.

- **On Router0 CLI:**

```
Router0(config-if)# ip route 192.168.20.0 255.255.255.0 10.0.0.2
Router0(config-if)# ip route 192.168.21.0 255.255.255.0 10.0.0.2
```

- **On Router1 CLI:**

```
Router1(config-if)# ip route 192.168.10.0 255.255.255.0 10.0.0.1
Router1(config-if)# ip route 192.168.11.0 255.255.255.0 10.0.0.1
```

2. Verify PC Gateways

On **PC0** (and PC1), make sure the **Default Gateway** is set to 192.168.10.1 (Router0's G0/0 IP). On **PC4** (and PC5), set the **Default Gateway** to 192.168.20.1 (Router1's G0/0 IP).

3. Test End-to-End

Open the Command Prompt on **PC0** and run:

```
ping 192.168.20.2
```

(Note: The first packet might say "Request timed out" due to ARP resolution across the routers, but subsequent packets will reply.)

Response: Looking at both routing tables, **neither router has static routes configured yet.**

Both tables only show C (directly connected) and L (local interface) entries. Without any S (static) entries, Router0 drops any traffic meant for 192.168.20.0 or 21.0, and Router1 drops traffic meant for 10.0 or 11.0.

Copy and paste these exact commands into each router's CLI to fix it:

1. Add Routes on Router0

(Run this in the Router0 CLI)

```
Router0# configure terminal
Router0(config)# ip route 192.168.20.0 255.255.255.0 10.0.0.2
Router0(config)# ip route 192.168.21.0 255.255.255.0 10.0.0.2
Router0(config)# end
```

2. Add Routes on Router1

(Run this in the Router1 CLI)

```
Router1# configure terminal
Router1(config)# ip route 192.168.10.0 255.255.255.0 10.0.0.1
Router1(config)# ip route 192.168.11.0 255.255.255.0 10.0.0.1
Router1(config)# end
```

How to Verify

Run `show ip route` again on either router. You should now see S lines for the remote subnets:

```
S    192.168.20.0/24 [1/0] via 10.0.0.2
S    192.168.21.0/24 [1/0] via 10.0.0.2
```

Once those static lines show up, ping from **PC0 to PC4** (e.g., ping 192.168.20.2). Remember to run the ping twice in a row if the first packet times out for ARP!

Response: The static routes on **Router0** are now configured correctly (S 192.168.20.0/24 and S 192.168.21.0/24 both show up via 10.0.0.2).

To finish the connection:

1. Apply Static Routes on Router1 Make sure you run the reverse routes on **Router1** CLI so it knows how to respond to Router0:

```
Router1# configure terminal
Router1(config)# ip route 192.168.10.0 255.255.255.0 10.0.0.1
Router1(config)# ip route 192.168.11.0 255.255.255.0 10.0.0.1
Router1(config)# end
```

2. Test from PC to PC

- Open **PC0** command prompt and run: ping 192.168.20.2 (PC4).
 - Run the ping command **twice** in a row. The first packet usually times out due to initial ARP resolution in Packet Tracer, but the rest will respond with Reply from 192.168.20.2.
-